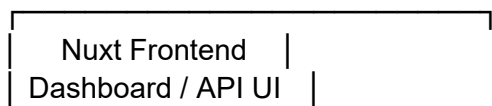


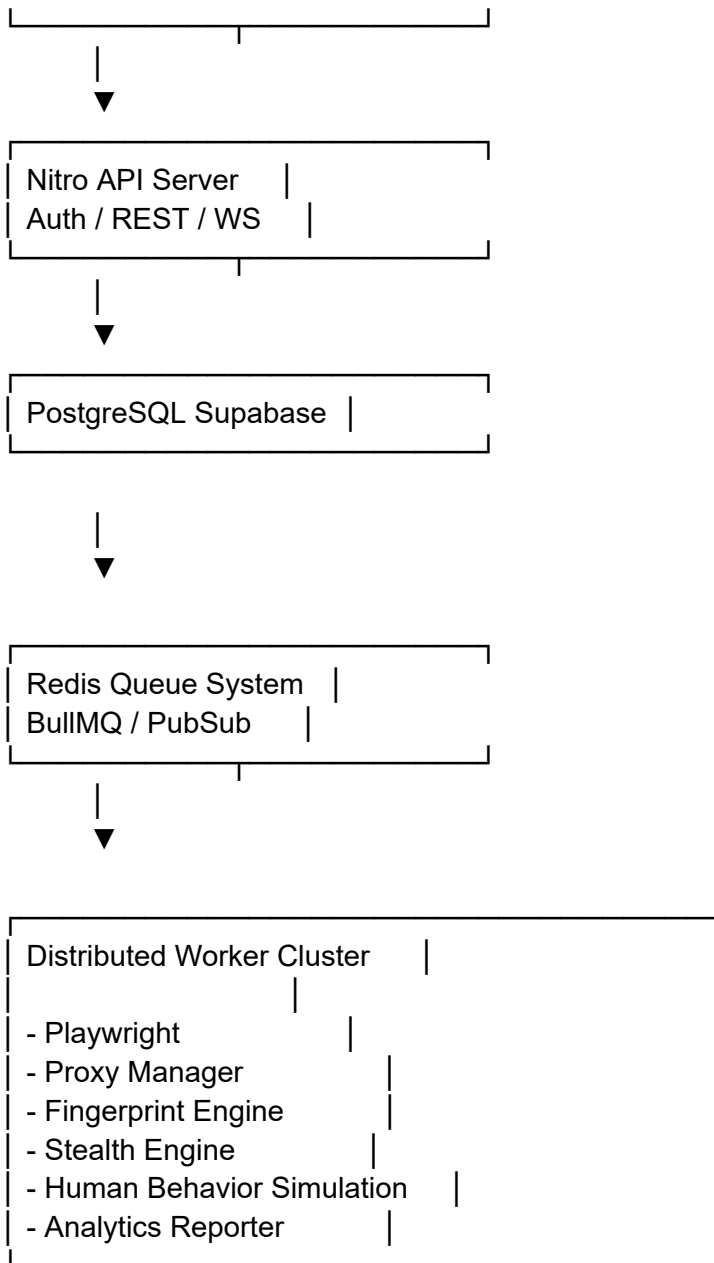
Modern Traffic Exchange Platform — Architecture & Feature Map

Core Technology Stack

Layer	Technology
Frontend	Nuxt 4
UI Components	Nuxt UI
Styling	Tailwind CSS
ORM	Prisma
Database	PostgreSQL (Supabase)
Auth	nuxt-auth-utils
Realtime	Nitro Native WebSocket
Queue	Redis
Upload Service	Cloudinary
Browser Automation	Playwright
Stealth Engine	playwright-stealth
Fingerprint Engine	playwright-with-fingerprints
Deployment	Docker Compose
Monitoring	Prometheus + Grafana
Logging	Loki / Pino
Scaling	Docker Swarm / Kubernetes Ready

High-Level System Architecture





Main System Modules

1. Dashboard App (Nuxt 4)

Features

Home Dashboard

- Stats Cards
 - Live Analytics
 - Campaign Charts
 - Active Sessions
 - Online Workers
 - Traffic Distribution Map
 - Browser Usage Stats
 - GEO Traffic Stats
-

Campaigns

- Create Campaign
 - Edit Campaign
 - Delete Campaign
 - Pause / Resume
 - Traffic Targeting
 - Device Targeting
 - GEO Targeting
 - Proxy Assignment
 - Duration Config
 - Behavior Pattern Config
-

Integrations

Per-user integrations:

- Residential Proxy
 - Mobile Proxy
 - SOCKS5
 - HTTP Proxy
 - Rotating Proxy
 - Multilogin
 - GoLogin
 - AdsPower
 - External Captcha Solver
-

Accounts

- User profile
- API key

- Subscription
 - Usage limits
 - Session management
-

Settings (Admin)

- Worker limits
 - Queue limits
 - Global proxy rules
 - GEO restrictions
 - Detection sensitivity
 - Ban policies
 - Rate limiting
-

2. Authentication System

Using:

[nuxt-auth-utils](#)

Providers

- Email Credential
 - OAuth Google
 - OAuth GitHub
-

3. Queue System

Redis + BullMQ

Responsibilities:

- campaign queue
- browser queue
- retry queue
- dead-letter queue

- analytics queue
-

Queue Types

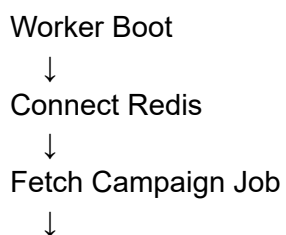
Queue	Function
campaign_queue	Traffic jobs
proxy_rotation_queue	Rotate proxy
analytics_queue	Store stats
behavior_queue	Human simulation
session_queue	Browser session
health_queue	Worker health
retry_queue	Retry failed jobs

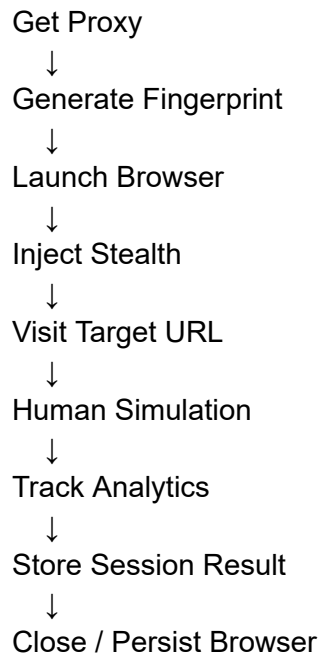
4. Worker Engine (Core)

Responsibilities

- Launch browser
 - Inject proxy
 - Generate fingerprint
 - Apply stealth mode
 - Simulate human behavior
 - Track analytics
 - Report completion
-

Worker Flow





5. Browser Pool System

Features

- Persistent sessions
 - Browser recycling
 - Session reuse
 - Context isolation
 - Cookie persistence
 - Fingerprint persistence
 - Multi browser engine
-

Browser Modes

Mode	Description
Ephemeral	One-time browser
Sticky Session	Reuse same identity
Persistent Profile	Long-lived session

6. Proxy Pool Manager

Supported

- HTTP
 - HTTPS
 - SOCKS5
 - Residential
 - Mobile
 - ISP Proxy
 - Rotating Proxy
-

Proxy Rotation Logic

Rotation Modes

Mode	Description
Per Session	Rotate every session
Sticky	Same IP for duration
Per Campaign	Shared campaign proxy
Smart Rotation	Detection-aware

Proxy Health System

Track:

- response time
- uptime
- block rate
- geo validation
- DNS leak

- blacklist detection
-

7. Fingerprint Engine

Using:

- [playwright-with-fingerprints](#)
 - Custom masking engine
-

Fingerprint Masking

Randomized Parameters

- userAgent
 - timezone
 - canvas
 - webgl
 - fonts
 - audioContext
 - navigator.webdriver
 - hardwareConcurrency
 - deviceMemory
 - geolocation
 - language
 - screen resolution
 - media devices
 - touch support
-

8. Playwright Stealth Engine

Using:

[playwright-stealth](#)

Additional Stealth Layer

Custom patches:

- webdriver removal
 - permission spoof
 - WebRTC masking
 - GPU spoofing
 - battery spoofing
 - plugin spoofing
 - speech voices spoofing
-

9. Human Behavior Engine

Simulated Activities

- random mouse movement
 - realistic scrolling
 - idle pause
 - random focus blur
 - tab switching
 - internal link clicks
 - keyboard typing
 - hover behavior
 - viewport resize
-

Advanced Human Simulation

AI-Based Timing Randomizer

Generate:

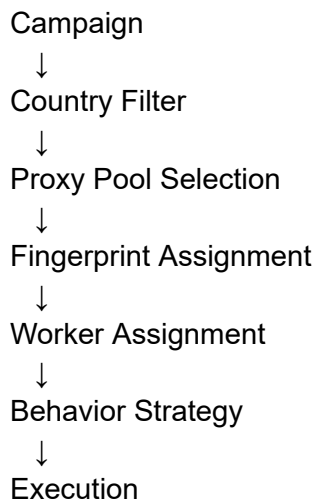
- natural reading time
 - attention span
 - engagement probability
-

10. Traffic Balancing Algorithm

Responsibilities

- distribute traffic evenly
 - avoid IP repetition
 - avoid fingerprint collision
 - GEO balancing
 - session balancing
 - worker balancing
-

Balancing Logic



11. Analytics Engine

Track Metrics

Traffic Metrics

- unique IP
- unique session
- total hits
- bounce rate
- average duration

- engagement rate
-

Device Metrics

- browser
 - OS
 - device type
 - screen size
 - timezone
-

GEO Metrics

- country
 - city
 - ISP
 - ASN
-

12. Realtime System

Using:

- Nitro Native WebSocket
 - Redis Pub/Sub
-

Live Events

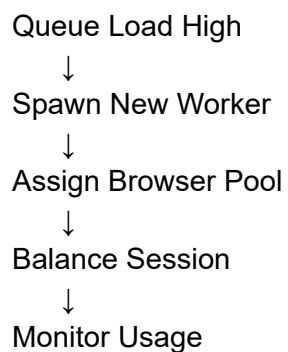
- live campaign stats
 - worker online/offline
 - queue updates
 - proxy health
 - browser status
 - realtime charts
-

13. Auto Scaling System

Capabilities

- worker auto spawn
 - auto restart
 - health recovery
 - queue balancing
 - memory monitoring
 - CPU monitoring
-

Scaling Strategy



14. GEO / Country Targeting

Target Modes

Mode	Description
Single Country	One GEO
Multiple Country	Multi GEO
Weighted GEO	% distribution
Dynamic GEO	Auto based proxy

15. Advanced Integrations

Optional Advanced

- Multilogin
 - GoLogin
 - AdsPower
 - CapMonster
 - 2Captcha
 - Turnstile Solver
-

16. Recommended Extra Features

A. Smart Detection Score

Internal risk scoring:

- fingerprint reuse
 - IP abuse
 - suspicious timing
 - proxy quality
-

B. Session Replay

Record:

- screenshots
 - click history
 - browser logs
-

C. Browser Crash Recovery

Restore:

- session
- cookies
- campaign progress

D. AI Fingerprint Generator

Generate:

- realistic combinations
 - device profile clustering
-

E. Heatmap Analytics

Track:

- mouse movement
 - click area
 - engagement zones
-

Database Main Tables

users
campaigns
campaign_targets
campaign_sessions
worker_nodes
browser_sessions
fingerprints
proxy_pools
proxy_logs
analytics_events
traffic_logs
integrations
queue_jobs
system_logs
geo_targets
behavior_profiles

Docker Architecture

Services

services:
 dashboard:
 api:
 postgres:
 redis:
 worker:
 orchestrator:
 analytics:
 prometheus:
 grafana:

Recommended Deployment

Phase 1 (MVP)

- Docker Compose
 - Single VPS
 - 5-20 workers
-

Phase 2 (Growth)

- Multi VPS
 - Distributed Redis
 - Centralized Queue
-

Phase 3 (Enterprise)

- Kubernetes
 - Auto scaling
 - GEO worker region
-

Security Recommendations

Important

- encrypted proxy credentials
 - session isolation
 - worker sandbox
 - rate limiting
 - API throttling
 - RBAC admin system
 - audit logs
 - worker authentication
-

Recommended Final Stack

Category	Recommendation
Frontend	Nuxt 4
UI	Nuxt UI
CSS	TailwindCSS
ORM	Prisma
DB	Supabase PostgreSQL
Queue	Redis + BullMQ
Realtime	Nitro WS
Worker	Playwright TS
Stealth	playwright-stealth
Fingerprint	playwright-with-fingerprints
Deployment	Docker
Monitoring	Grafana
Scaling	Kubernetes Ready